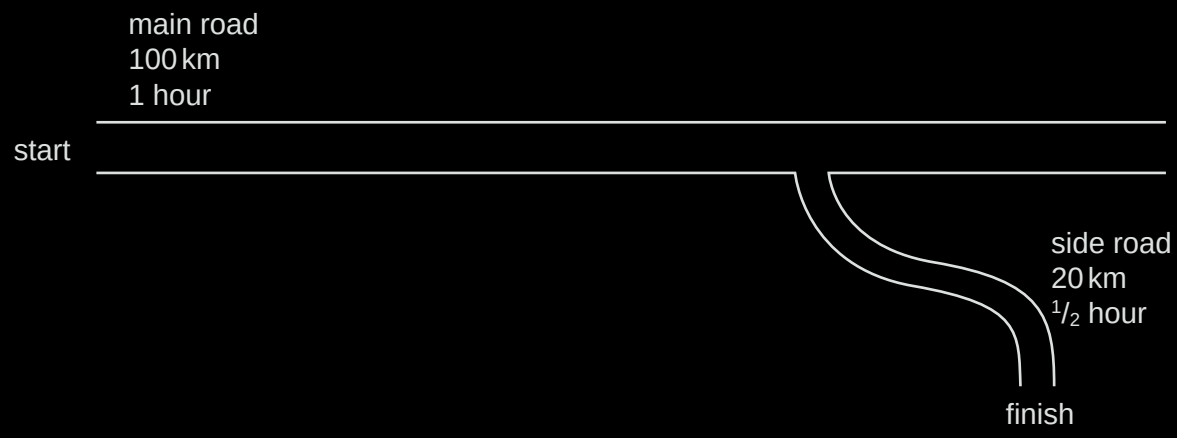


3 A car takes 1 hour to travel 100 km along a main road and then ½ hour to travel 20 km along a side road.



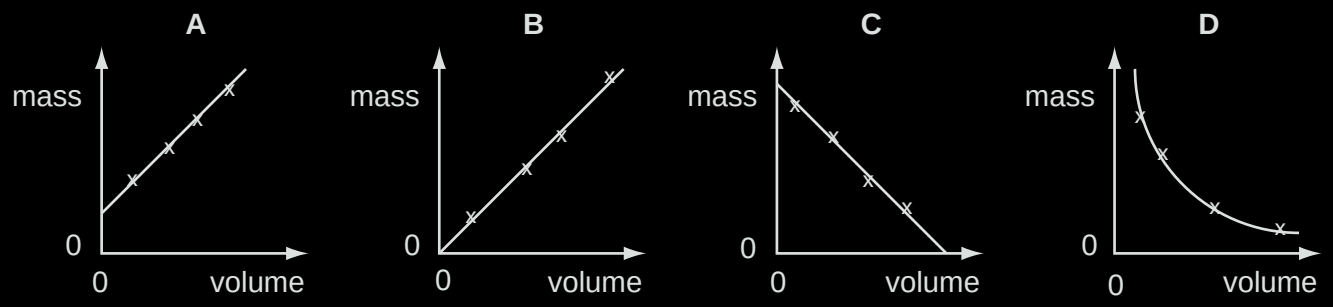
What is the average speed of the car for the whole journey?

A 60 km / h

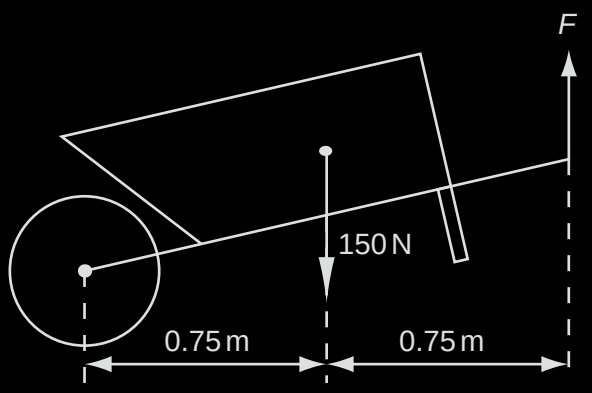
B 70 km / h

C 80 km / h

D 100 km / h

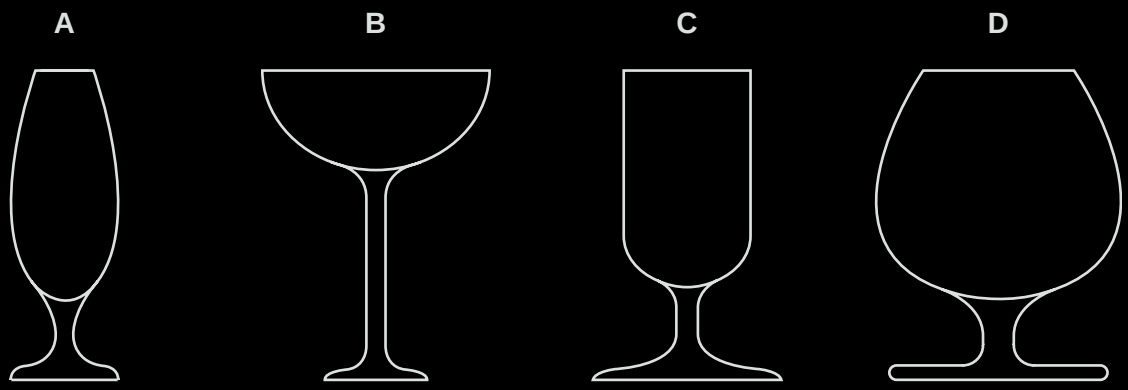


6 The diagram shows a wheelbarrow and its load, which have a total weight of 150 N. This is supported by a vertical force *F* at the ends of the handles.



What is the value of *F*?

A 75 N **B** 150 N **C** 225 N **D** 300 N



8 The table shows how the extension of a spring varies with load.

load / N	0	2	4	6	8	10	12	14	16
extension / cm	0	3	6	9	12	15	20	27	38

Between which two loads would you find the limit of proportionality?

A 0 N and 2 N

B 8 N and 10 N

C 10 N and 12 N

D 14 N and 16 N

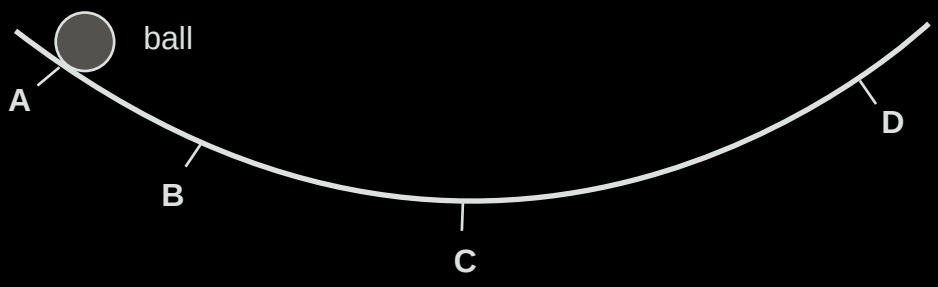
9 Which is the correct statement about force and velocity?

A Force and velocity are both scalars.

B Force and velocity are both vectors.

C Force is a scalar, velocity is a vector.

D Force is a vector, velocity is a scalar.



11 The speed of light is 3×10^8 m/s.

What is the energy equivalent of 2 kg of matter?

A $2 \times 3 \times 10^8$ J

B $2000 \times 3 \times 10^8$ J

C $\frac{1}{2} \times 2 \times (3 \times 10^8)^2$ J

D $2 \times (3 \times 10^8)^2$ J

12 What does **not** affect the pressure at a point beneath the surface of a liquid?

A area of the liquid surface

B density of the liquid

C depth of the point below the surface

D strength of the gravitational field

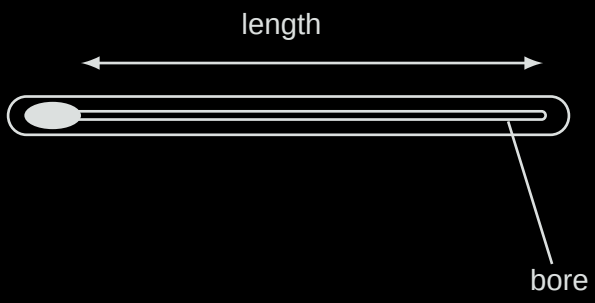
13 A small table weighing 40 N stands on four legs, each having an area of 0.001 m^2.

What is the pressure of the table on the floor?

A 400 N / m^2 **B** 1000 N / m^2 **C** 10 000 N / m^2 **D** 40 000 N / m^2

14 Which factors increase the rate of evaporation of a liquid?

	increasing its temperature	increasing its surface area	increasing its depth
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes



Which will have the greatest sensitivity?

A 10 cm long and bore 0.75 mm wide

B 15 cm long and bore 0.50 mm wide

C 25 cm long and bore 0.10 mm wide

D 30 cm long and bore 0.25 mm wide

16 In an experiment to find the specific heat capacity of a metal, it is found that 5200 J is needed to raise the temperature of a 2 kg block by 20 °C.

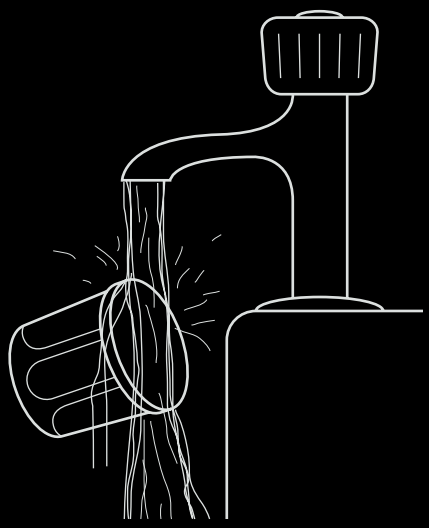
What value for the specific heat capacity is given by these results?

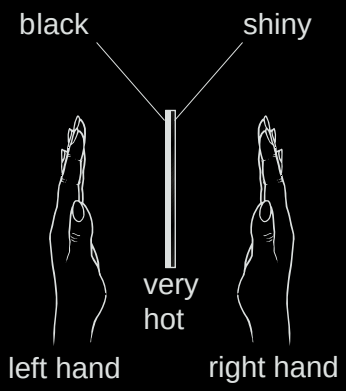
A 130 J / (kg °C)

B 520 J / (kg °C)

C 52 000 J / (kg °C)

D 104 000 J / (kg °C)





Her left hand feels warmer than her right hand.

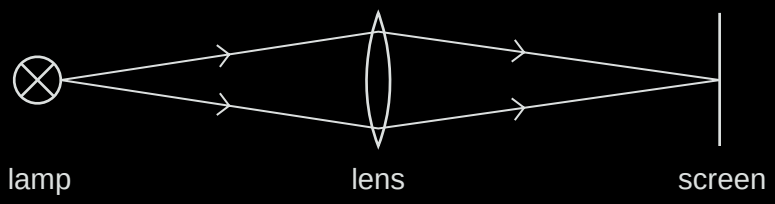
Which statement is the correct conclusion from the experiment?

A The black side is hotter than the shiny one.

B The black side radiates more heat.

C The shiny side radiates more heat.

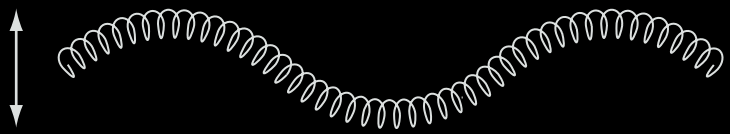
D The shiny side is cooling down faster than the black side.



B a spring pushed backwards and forwards



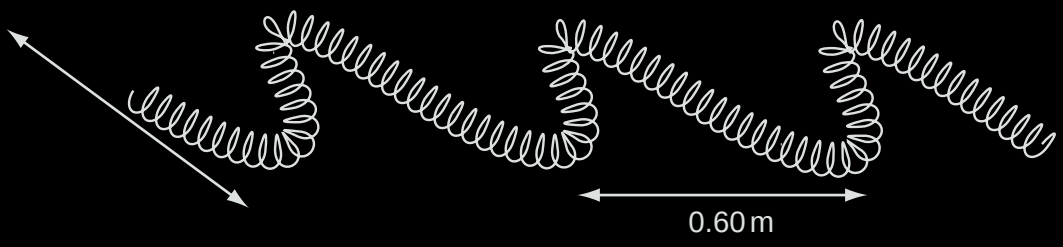
C a spring pushed up and down



D a water ripple caused by a dipper moving up and down



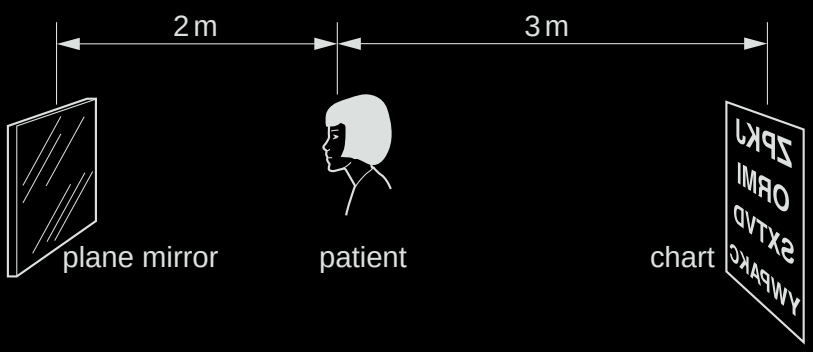
20 The diagram shows part of a spring that is shaken from side to side to produce a wave.



The distance between successive peaks is 0.60 m and the frequency is 2.5 Hz.

How long does it take for a wave to travel 3.0 m along the spring?

A 0.20 s **B** 0.50 s **C** 2.0 s **D** 5.0 s

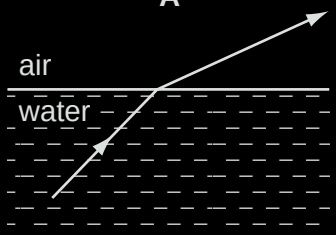


How far away from the patient is the image of the chart?

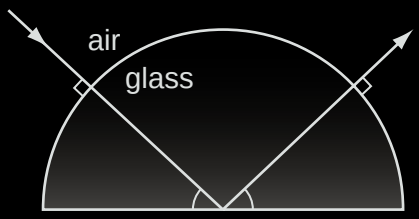
A 2 m **B** 4 m **C** 5 m **D** 7 m

22 In which diagram is the path of the light ray **not** correct?

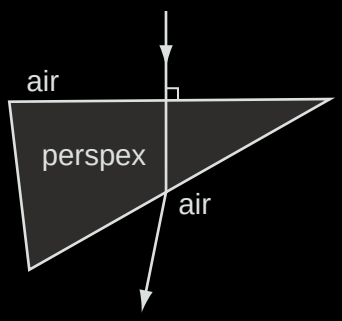
A



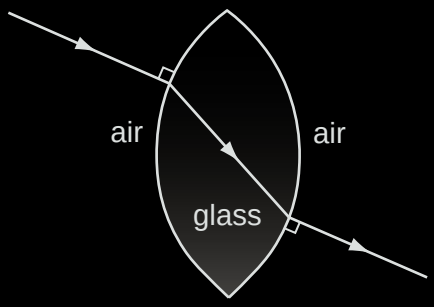
B



C



D



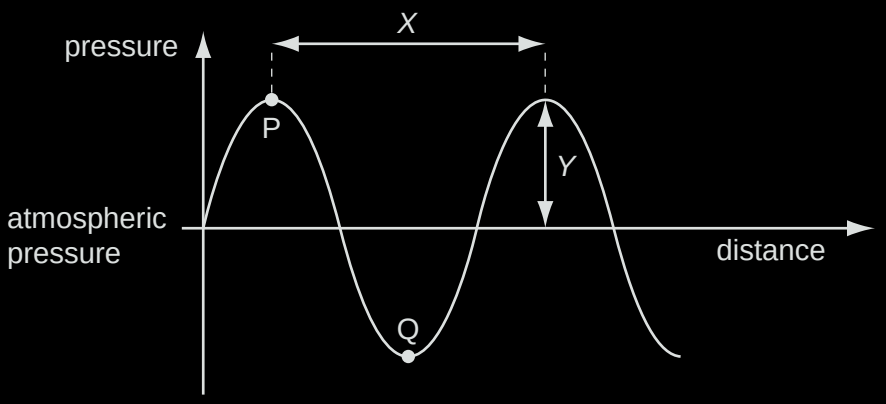
23 Which statement is true for all electromagnetic waves?

A They are longitudinal.

B They can be seen.

C They have the same frequency in air.

D They travel at the same speed in a vacuum.

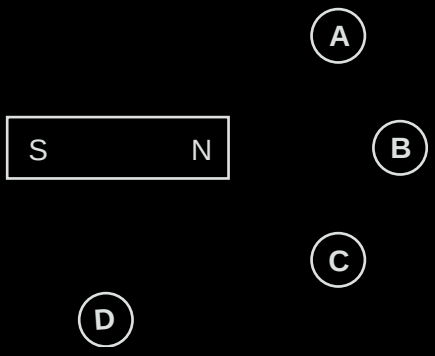


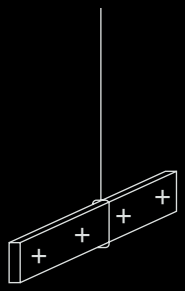
Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction at	wavelength is
A	P	*X*
B	P	*Y*
C	Q	*X*
D	Q	*Y*

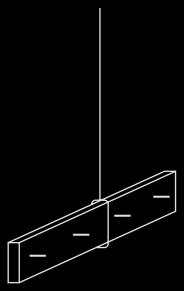
25 Four plotting compasses are placed near a bar magnet. You may ignore any effects of the Earth's magnetic field.

In which position does the compass appear like this,  ?





acetate strip



polythene strip

Two rods **X** and **Y** are brought up in turn to these two strips. Rod **X** attracts the acetate strip but repels the polythene strip. Rod **Y** does not repel either the acetate strip or the polythene strip.

Which type of charge is on each rod?

	rod **X**	rod **Y**
A	negative	positive
B	negative	uncharged
C	positive	negative
D	positive	uncharged

27 A cell is connected to a resistor.

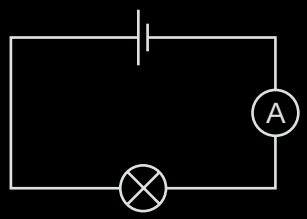
What is the e.m.f. of the cell equal to?

A The potential difference across the resistor for each unit of current.

B The power produced in the circuit for each unit of charge that passes.

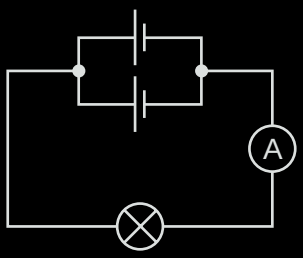
C The work done in the circuit for each unit of charge that passes.

D The work done in the circuit for each unit of current.

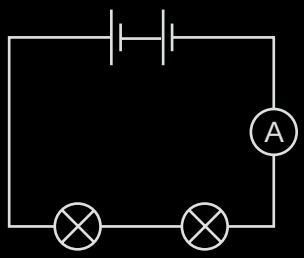


In which circuit, using identical cells, lamps and ammeters, is the current reading 2 A?

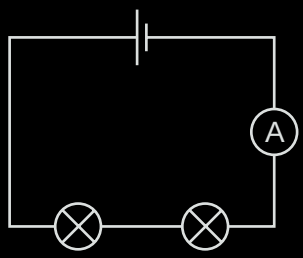
A



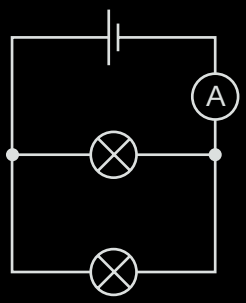
B



C

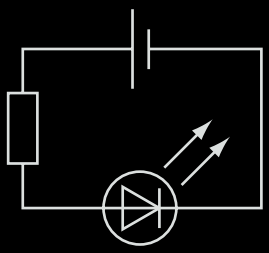


D

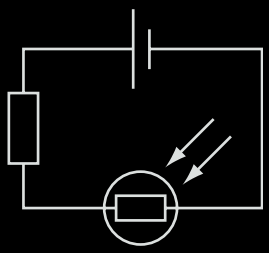


29 Which circuit contains a cell, a light-emitting diode and a fixed resistor?

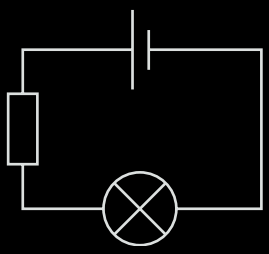
A



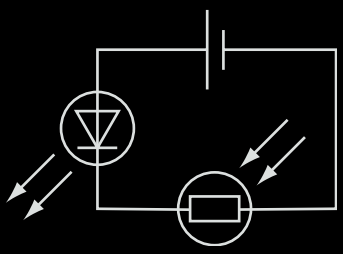
B



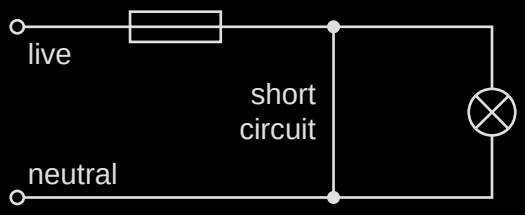
C



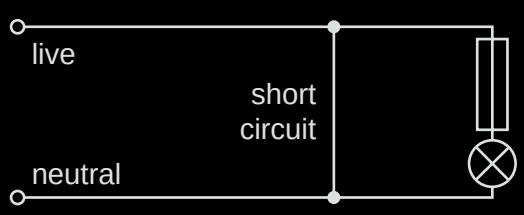
D



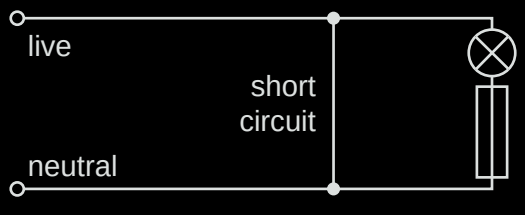
A



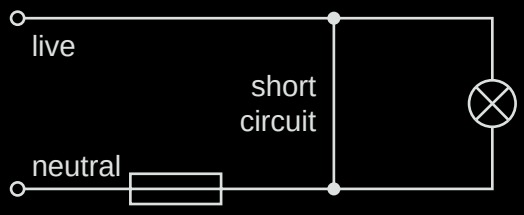
B



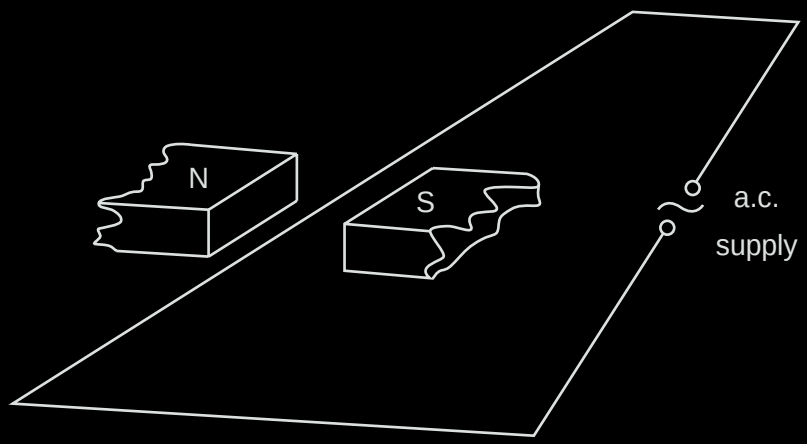
C



D



31 An alternating current is passed through a wire stretched between the poles of a magnet.



Which way will the wire move?

A left and right

B right only

C up and down

D up only

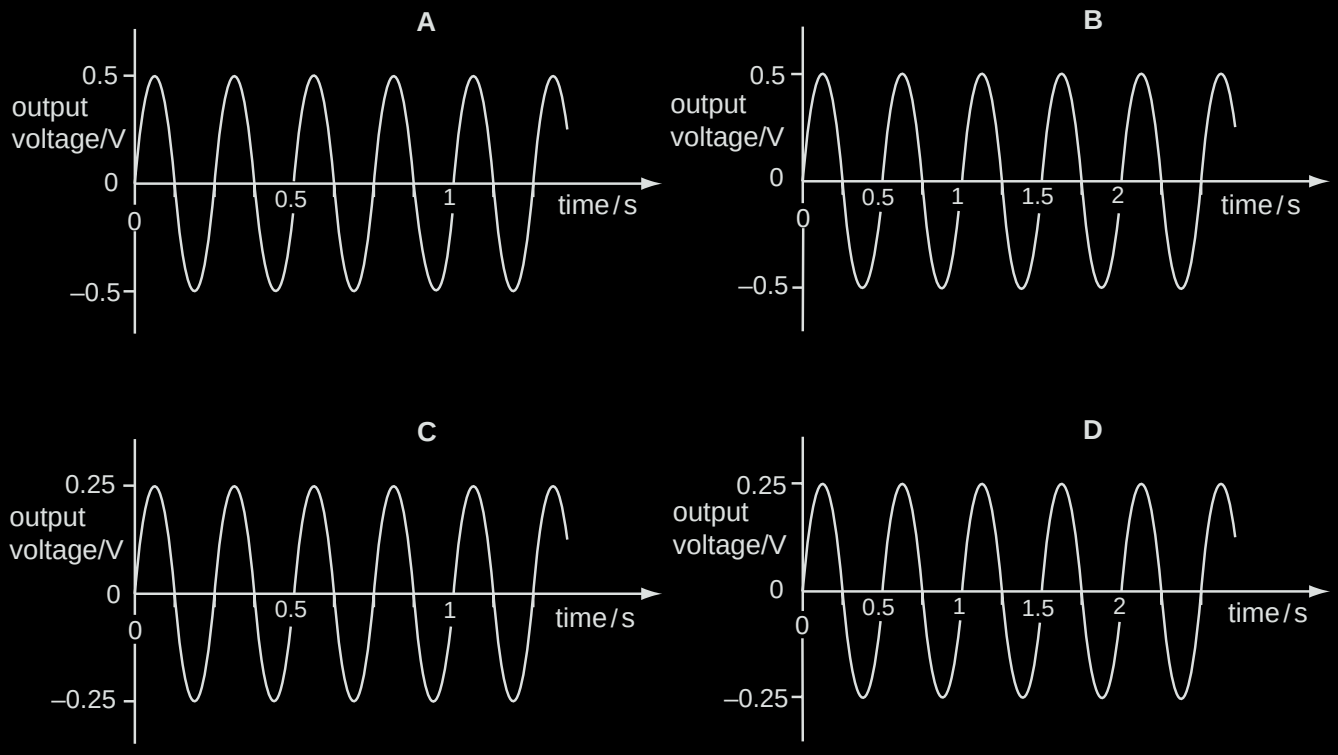


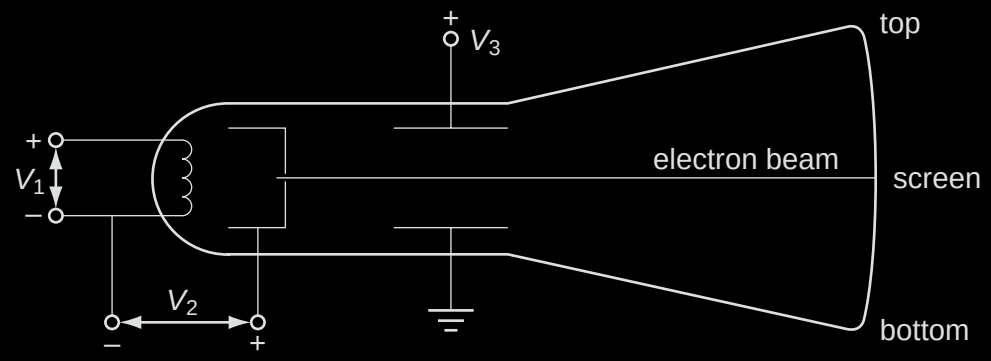
What is true about the direction and size of the currents?

	direction of currents	size of currents
A	same	larger in **X** than in **Y**
B	same	smaller in **X** than in **Y**
C	different	larger in **X** than in **Y**
D	different	smaller in **X** than in **Y**

33 A girl turns the handle of a small a.c. generator four times each second. The generator produces a maximum output voltage of 0.5 V.

Which of the following graphs best shows this?





Which change in voltage results in the position of the spot on the screen moving towards the top?

A decreasing V_1

B decreasing V_3

C increasing V_2

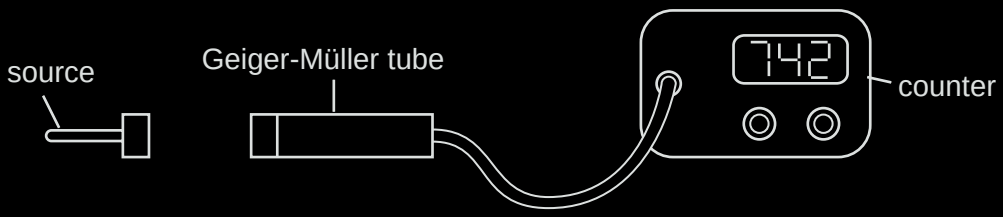
D increasing V_3



What is the temperature of the thermistor when the voltmeter reads 2 V?

A 20 °C **B** 40 °C **C** 60 °C **D** 80 °C

38 A radioactive source is placed 3 cm from a Geiger-Müller tube in air. The average count rate is 742 counts / minute.



Then, in three experiments, measurements are taken with sheets of different materials placed between the source and the tube. The results are recorded in the table.

material of sheet between source and tube	thin card	aluminium foil	thick lead
average count rate / counts per minute	273	275	68

Which types of radiation does the source emit?

A alpha and beta only

B alpha and gamma only

C beta and gamma only

D alpha, beta and gamma